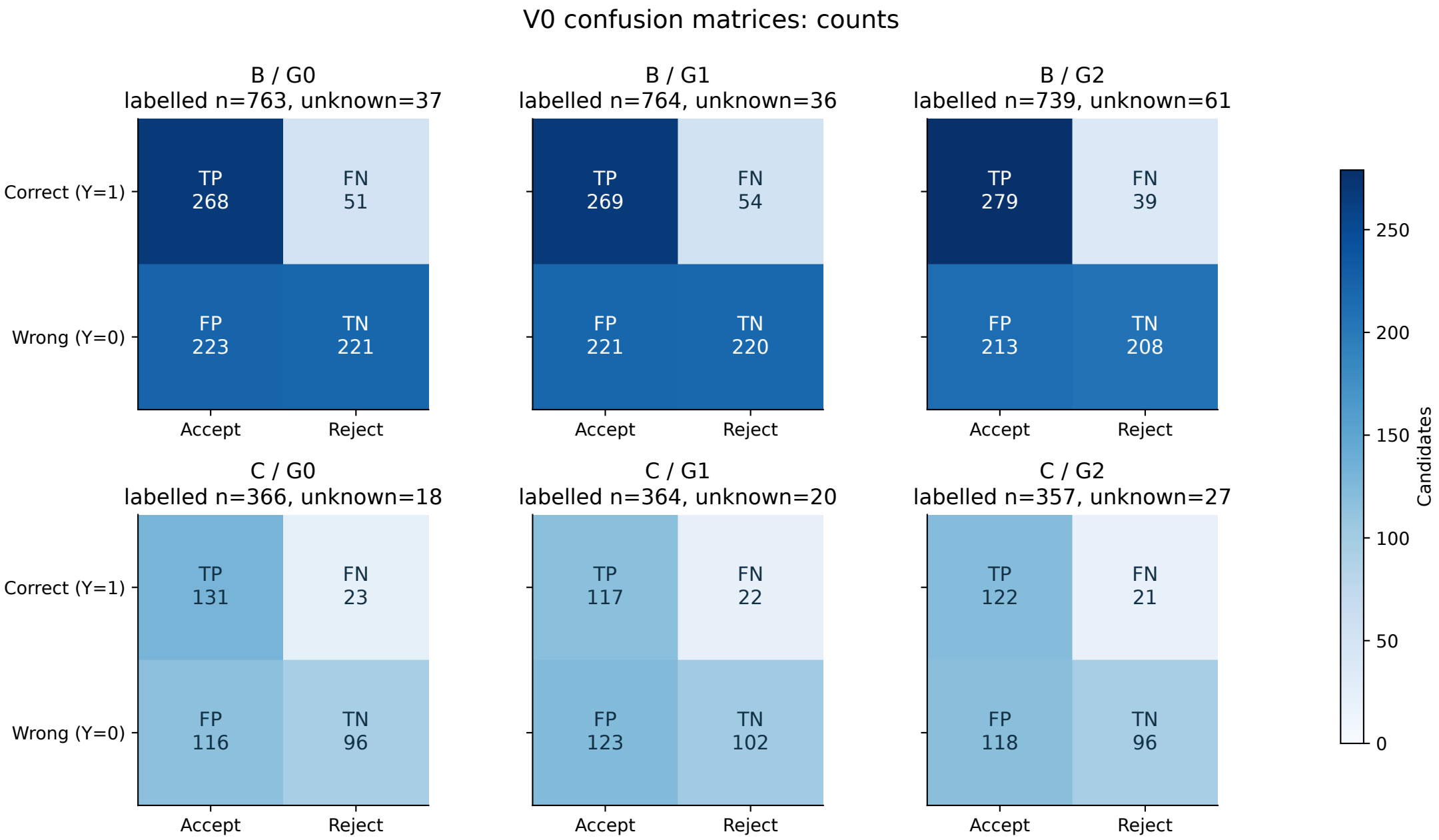
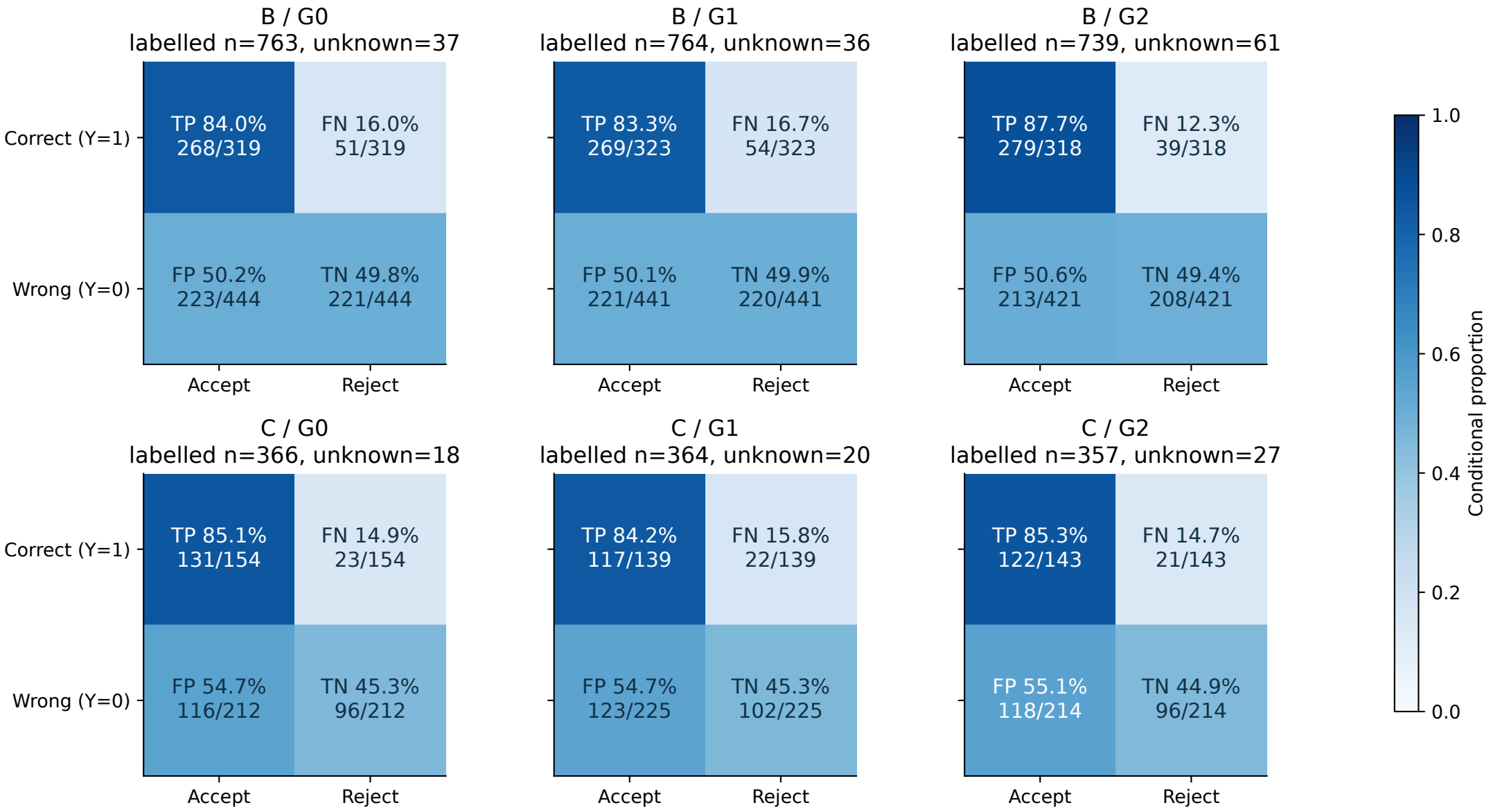
Fixed V0 across generator updates

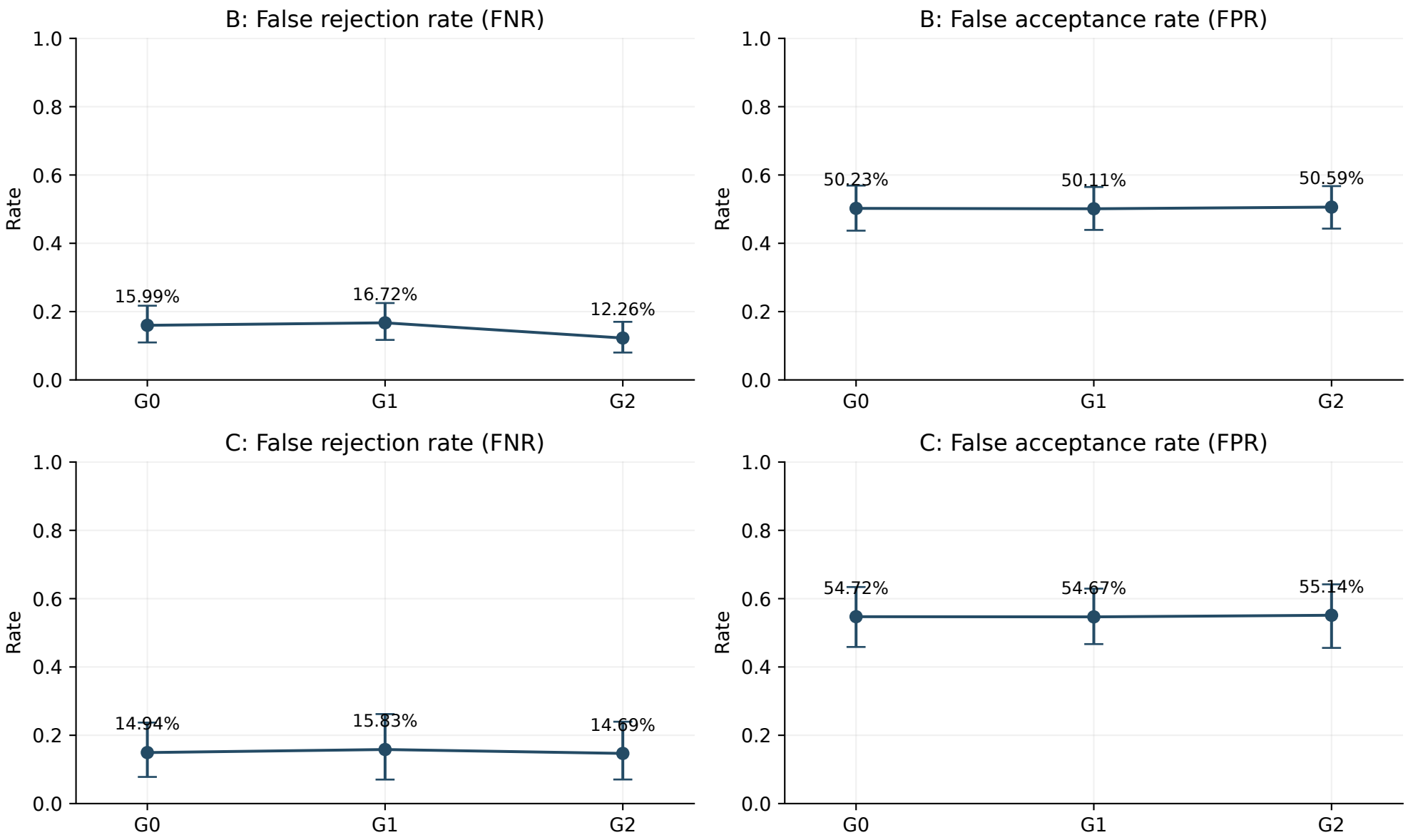
DAPO internal splits: B = 200 questions; C = 96 eligible questions
4 candidates/question/round. G0 -> G1_SFT -> G2_SFT. Threshold = 0.5.

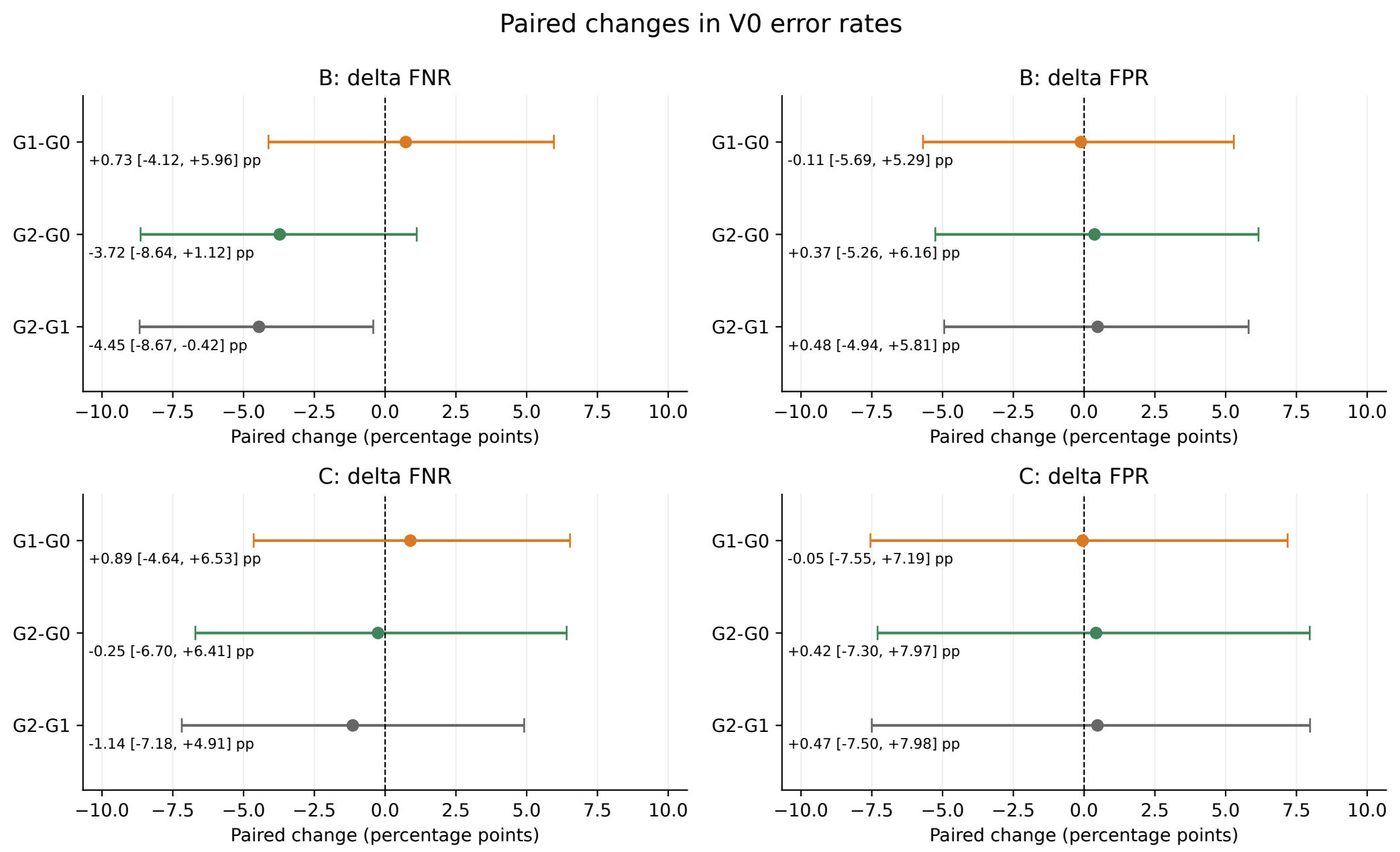
Split / G	FNR	FPR	Balanced acc.	ROC AUC	Unknown
B / G0	15.99%	50.23%	66.89%	0.763	37/800
B / G1	16.72%	50.11%	66.58%	0.764	36/800
B / G2	12.26%	50.59%	68.57%	0.789	61/800
C / G0	14.94%	54.72%	65.17%	0.766	18/384
C / G1	15.83%	54.67%	64.75%	0.788	20/384
C / G2	14.69%	55.14%	65.09%	0.796	27/384

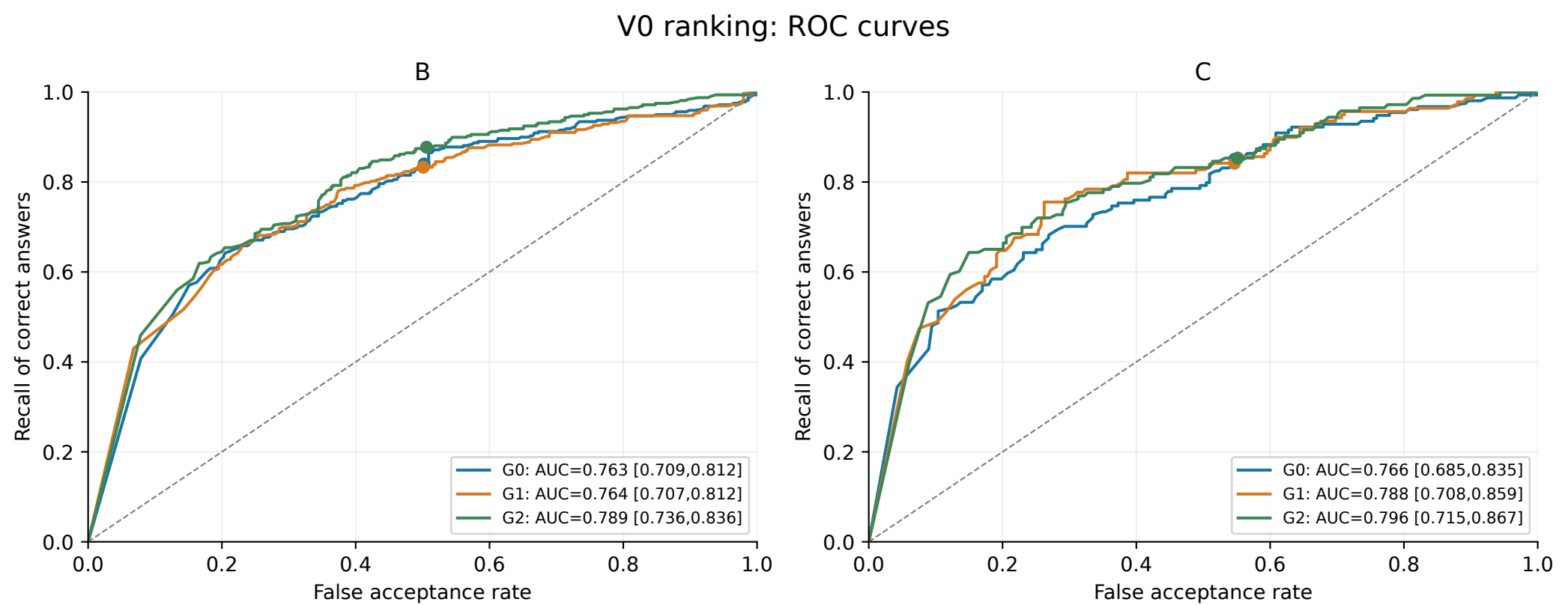
Primary drift: paired changes in FNR/FPR relative to G0.
95% question-cluster percentile bootstrap; 2,000 replicates.
B/C diagnostics only. T remains sealed. No new inference or training.


V0 confusion matrices: counts
B / G0
labelled n=763, unknown=37
TP
268
FN
51
FP
223
TN
221
B / G1
labelled n=764, unknown=36
TP
269
FN
54
FP
221
TN
220
B / G2
labelled n=739, unknown=61
TP
279
FN
39
FP
213
TN
208
C / G0
labelled n=366, unknown=18
TP
131
FN
23
FP
116
TN
96
C / G1
labelled n=364, unknown=20
TP
117
FN
22
FP
123
TN
102
C / G2
labelled n=357, unknown=27
TP
122
FN
21
FP
118
TN
96
Correct (Y=1)
Wrong (Y=0)
Accept
Reject
Candidates
250
200
150
100
50
0
Positive = independently correct final answer. Unknown labels excluded. FN/top-right = false rejection; FP/bottom-left = false acceptance.

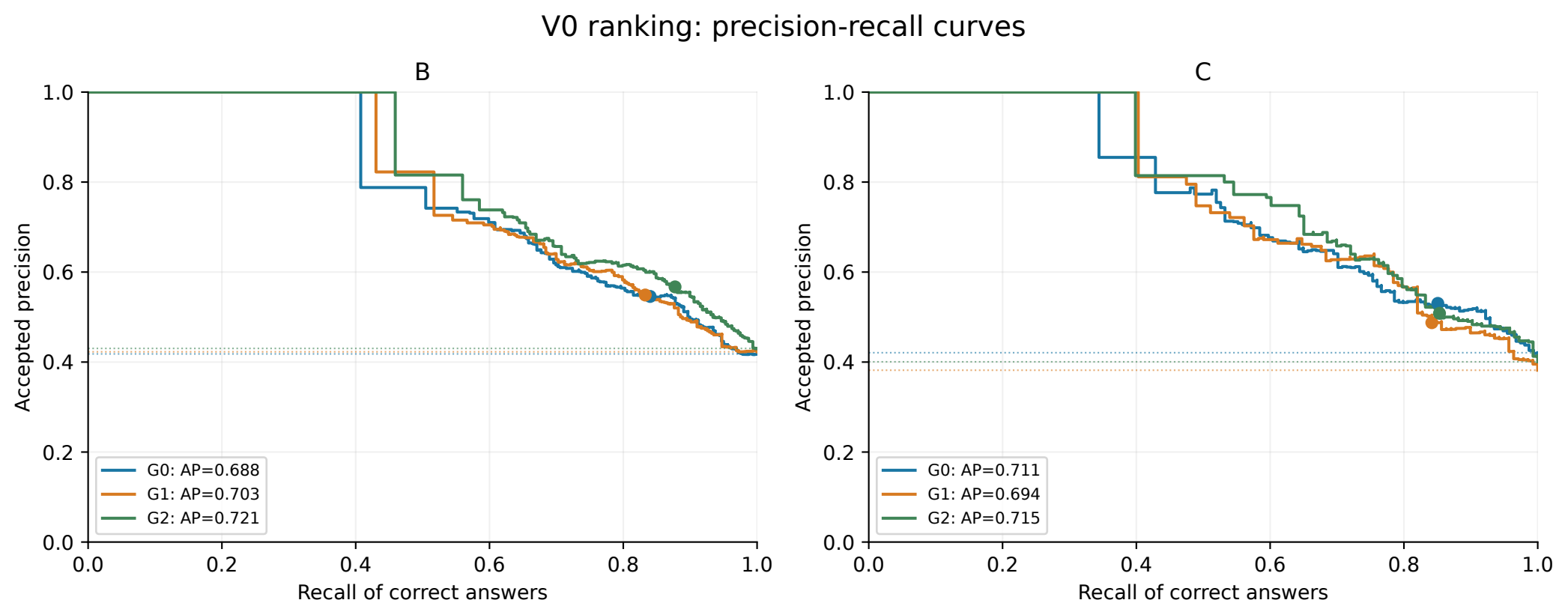

V0 confusion matrices: row-normalized rates
B / G0
labelled n=763, unknown=37
TP 84.0%
268/319
FN 16.0%
51/319
FP 50.2%
223/444
TN 49.8%
221/444
B / G1
labelled n=764, unknown=36
TP 83.3%
269/323
FN 16.7%
54/323
FP 50.1%
221/441
TN 49.9%
220/441
B / G2
labelled n=739, unknown=61
TP 87.7%
279/318
FN 12.3%
39/318
FP 50.6%
213/421
TN 49.4%
208/421
C / G0
labelled n=366, unknown=18
TP 85.1%
131/154
FN 14.9%
23/154
FP 54.7%
116/212
TN 45.3%
96/212
C / G1
labelled n=364, unknown=20
TP 84.2%
117/139
FN 15.8%
22/139
FP 54.7%
123/225
TN 45.3%
102/225
C / G2
labelled n=357, unknown=27
TP 85.3%
122/143
FN 14.7%
21/143
FP 55.1%
118/214
TN 44.9%
96/214
Correct (Y=1)
Wrong (Y=0)
Accept
Reject
Conditional proportion
1.0
0.8
0.6
0.4
0.2
0.0
Positive = independently correct final answer. Unknown labels excluded. FN/top-right = false rejection; FP/bottom-left = false acceptance.


Fixed V0 error rates across G rounds
B: False rejection rate (FNR)
B: False acceptance rate (FPR)
C: False rejection rate (FNR)
C: False acceptance rate (FPR)
Rate
1.0
0.8
0.6
0.4
0.2
0.0
G0
G1
G2
15.99%
16.72%
12.26%
50.23%
50.11%
50.59%
14.94%
15.83%
14.69%
54.72%
54.67%
55.14%
Bars: 95% question-cluster bootstrap intervals; 2,000 replicates. Same 0-1 y scale; V0 and threshold are fixed.

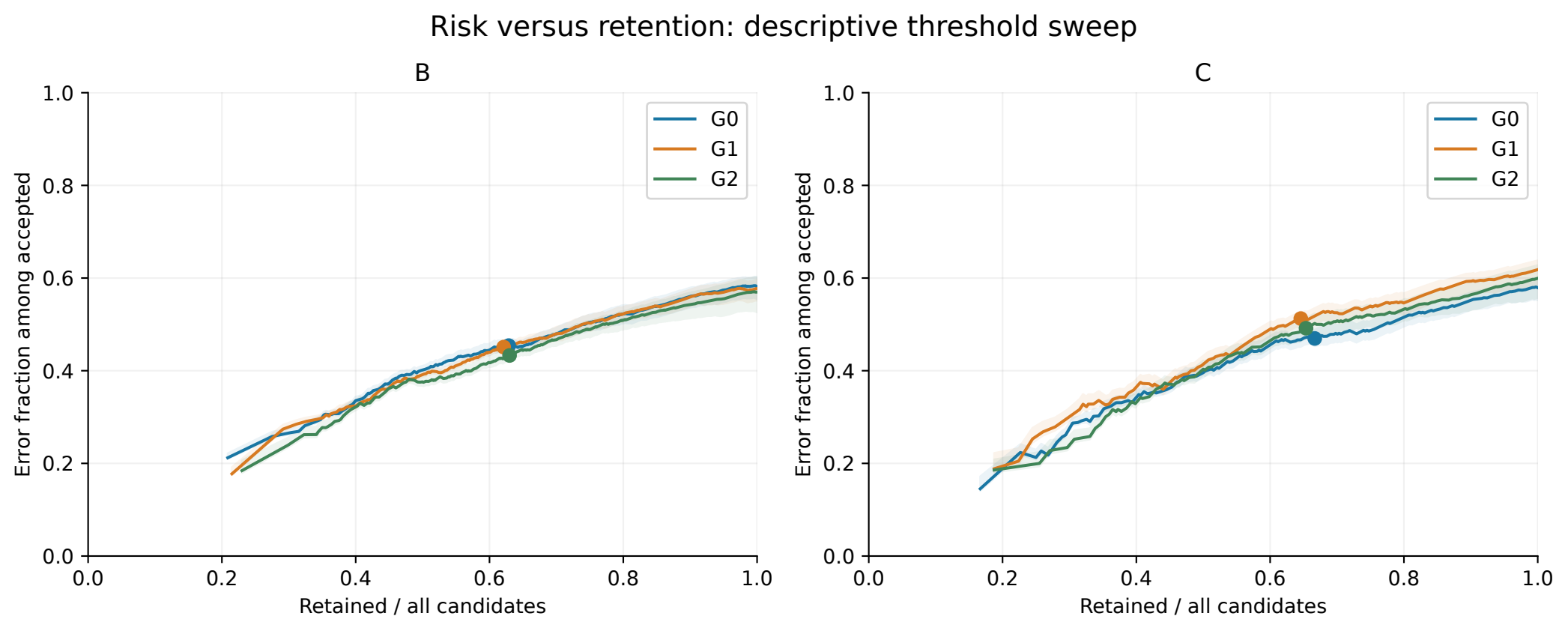

Paired changes in V0 error rates
B: delta FNR
G1-G0
+0.73 [-4.12, +5.96] pp
G2-G0
-3.72 [-8.64, +1.12] pp
G2-G1
-4.45 [-8.67, -0.42] pp
−10.0 −7.5 −5.0 −2.5 0.0 2.5 5.0 7.5 10.0
Paired change (percentage points)
B: delta FPR
G1-G0
-0.11 [-5.69, +5.29] pp
G2-G0
+0.37 [-5.26, +6.16] pp
G2-G1
+0.48 [-4.94, +5.81] pp
−10.0 −7.5 −5.0 −2.5 0.0 2.5 5.0 7.5 10.0
Paired change (percentage points)
C: delta FNR
G1-G0
+0.89 [-4.64, +6.53] pp
G2-G0
-0.25 [-6.70, +6.41] pp
G2-G1
-1.14 [-7.18, +4.91] pp
−10.0 −7.5 −5.0 −2.5 0.0 2.5 5.0 7.5 10.0
Paired change (percentage points)
C: delta FPR
G1-G0
-0.05 [-7.55, +7.19] pp
G2-G0
+0.42 [-7.30, +7.97] pp
G2-G1
+0.47 [-7.50, +7.98] pp
−10.0 −7.5 −5.0 −2.5 0.0 2.5 5.0 7.5 10.0
Paired change (percentage points)
Same question clusters resampled across rounds. Pointwise 95% intervals; exploratory, unadjusted. Crossing zero does not establish equivalence.

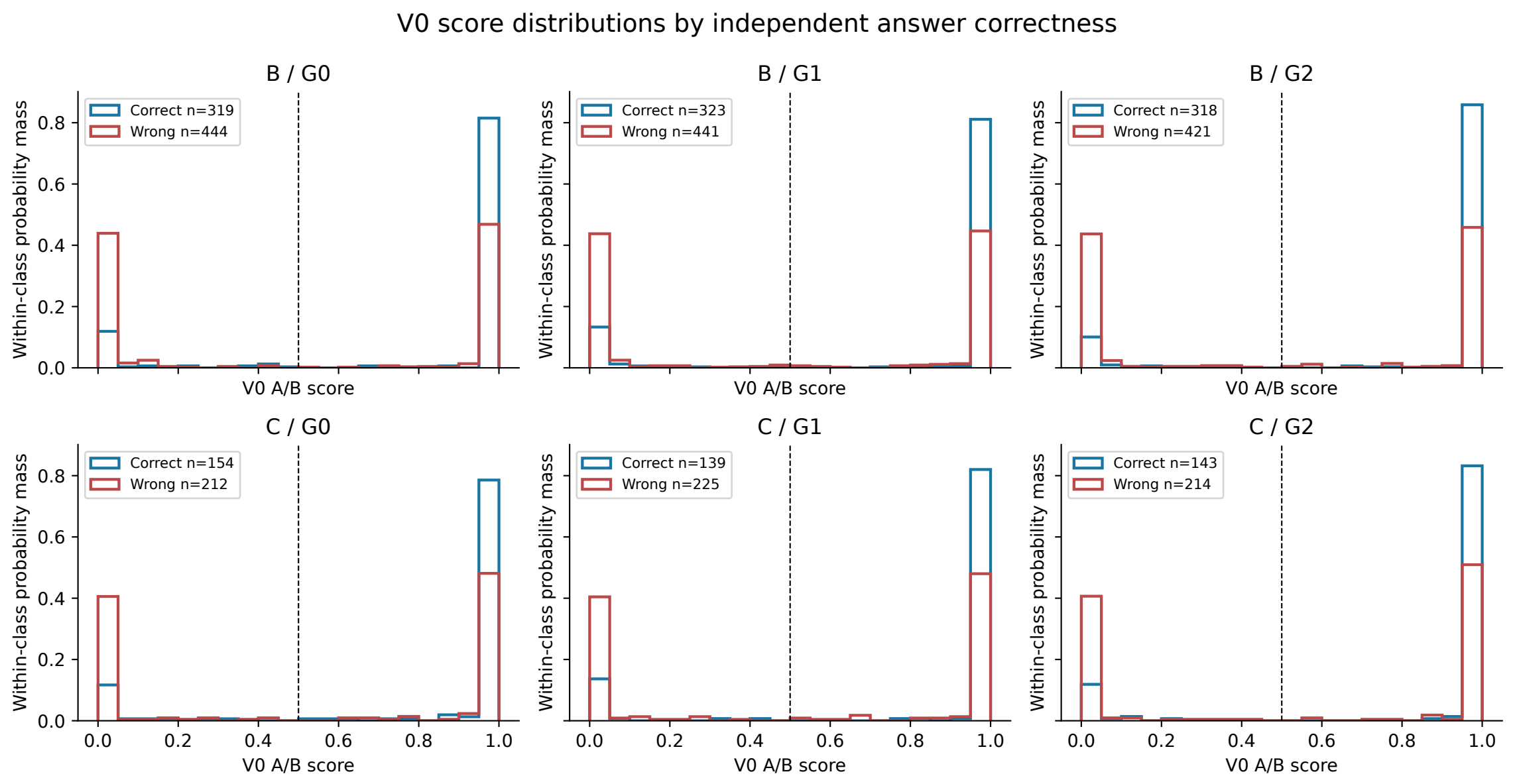


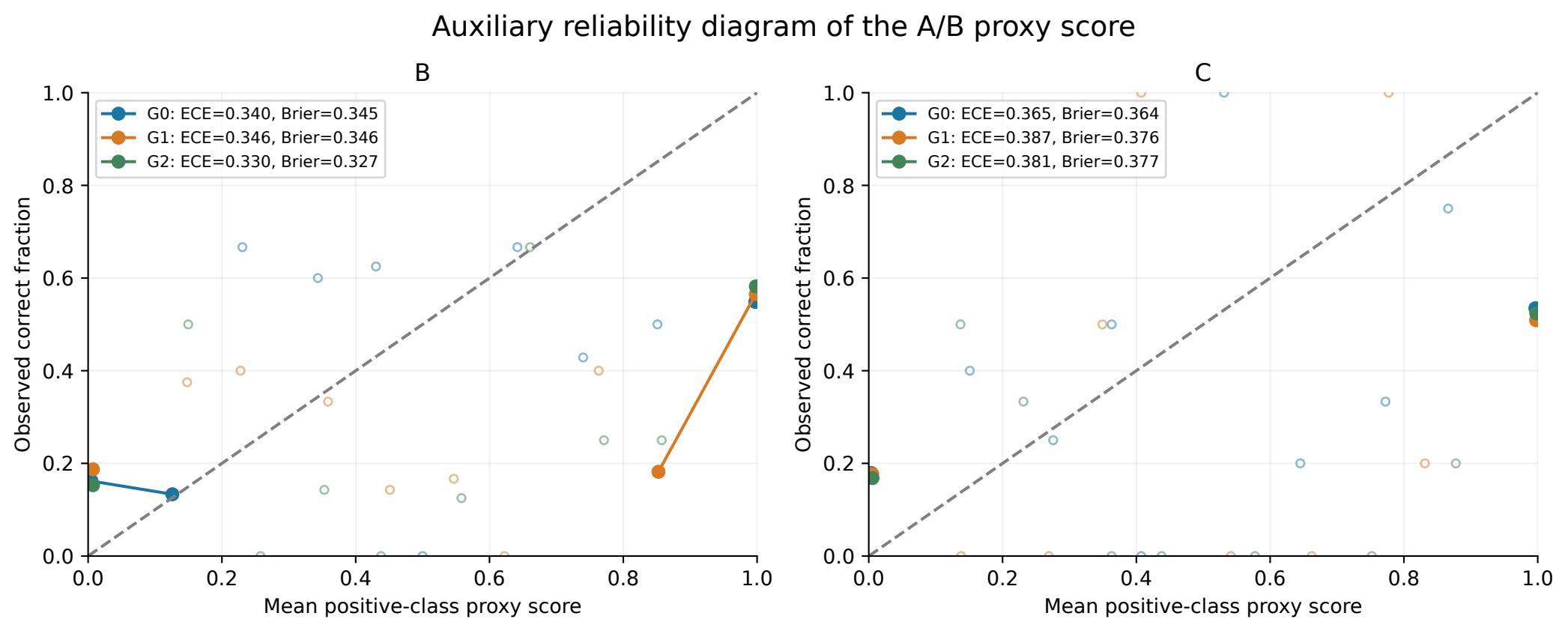
Verifiable subset only. Dots: frozen threshold 0.5. PR dotted lines: per-round correct-answer prevalence. AP is noninterpolated average precision.

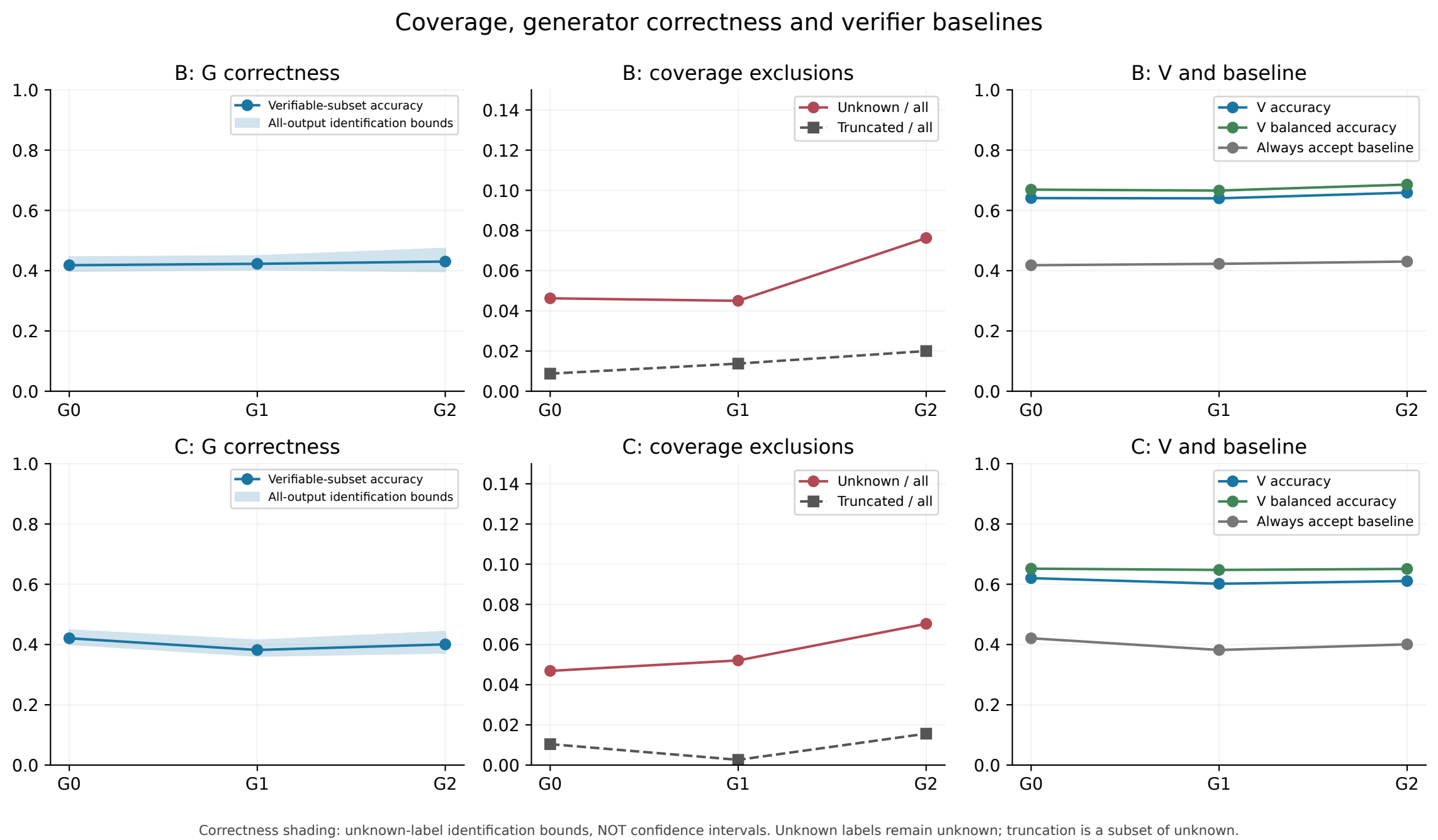


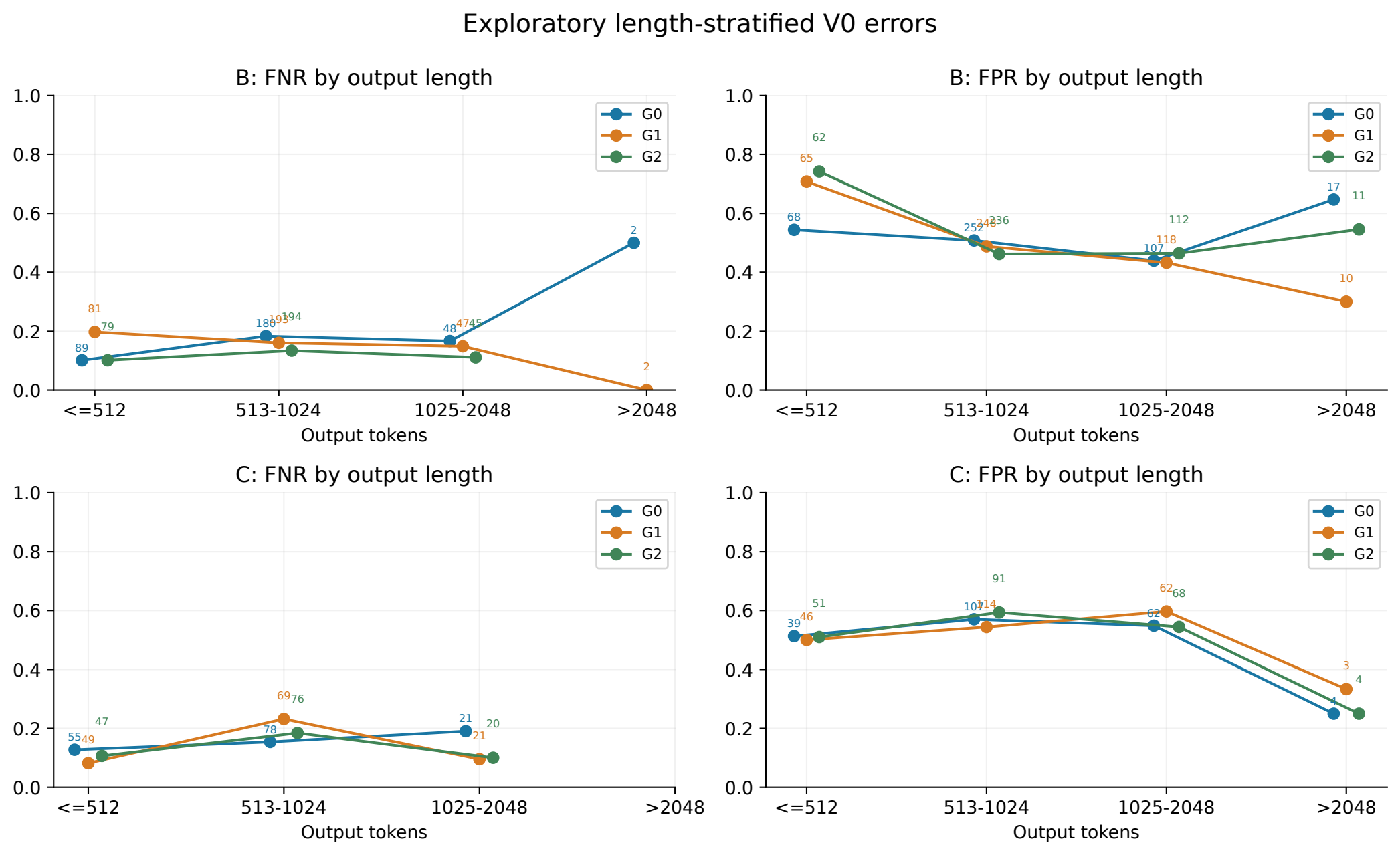
Verifiable subset only. Dots: frozen threshold 0.5. PR dotted lines: per-round correct-answer prevalence. AP is noninterpolated average precision.

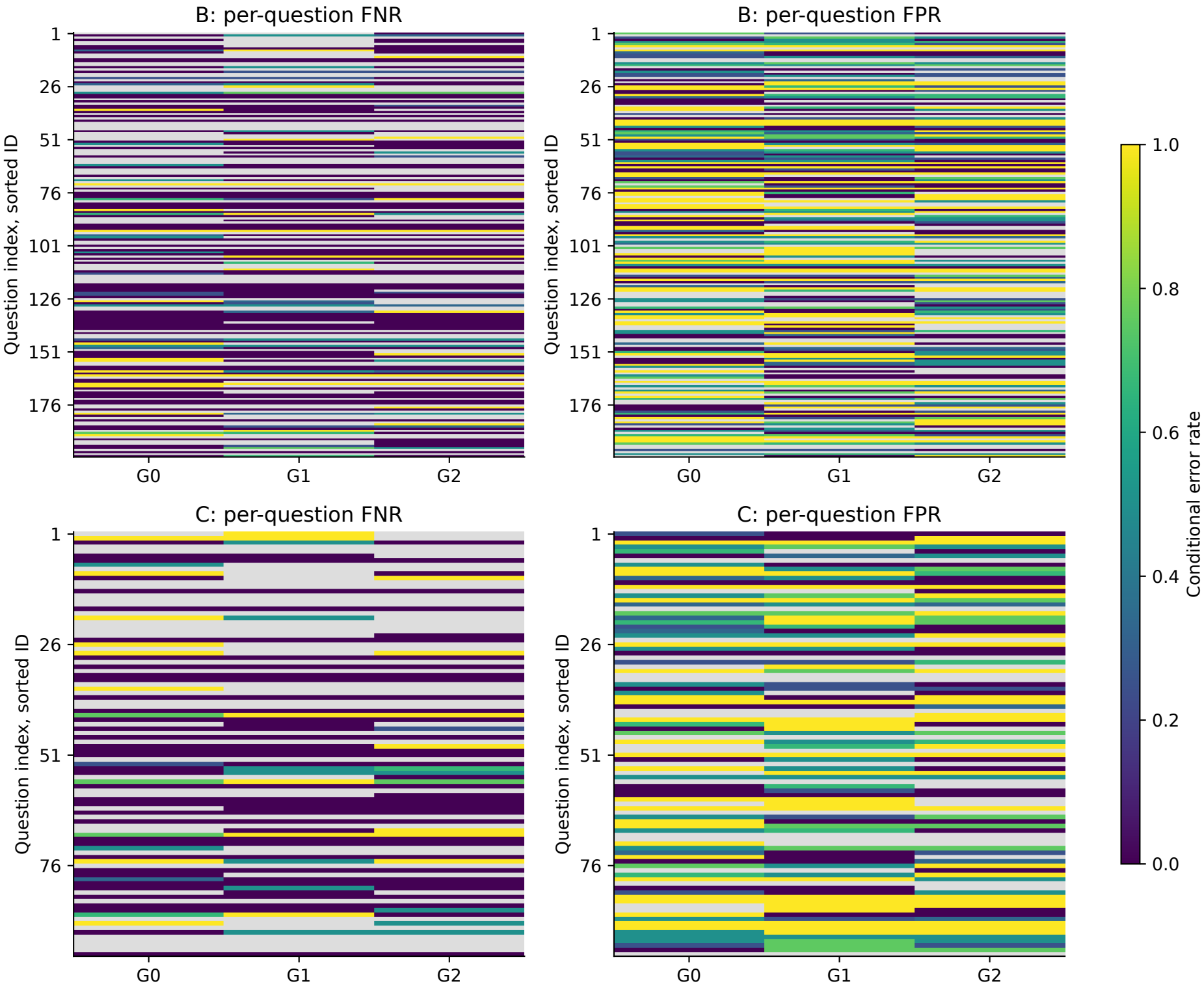

Risk versus retention: descriptive threshold sweep
B
C
G0
G1
G2
Error fraction among accepted
Retained / all candidates
0.0
0.2
0.4
0.6
0.8
1.0
Lines: error rate in labelled accepted subset. Bands: unknown-label identification bounds over ALL accepted. Dots: unchanged threshold 0.5.


V0 score distributions by independent answer correctness
B / G0
Correct n=319
Wrong n=444
B / G1
Correct n=323
Wrong n=441
B / G2
Correct n=318
Wrong n=421
C / G0
Correct n=154
Wrong n=212
C / G1
Correct n=139
Wrong n=225
C / G2
Correct n=143
Wrong n=214
Within-class probability mass
V0 A/B score
0.0
0.2
0.4
0.6
0.8
1.0
Each class is normalized separately. Score is a conditional-label proxy, not an established calibrated probability.


Auxiliary reliability diagram of the A/B proxy score
B
C
G0: ECE=0.340, Brier=0.345
G1: ECE=0.346, Brier=0.346
G2: ECE=0.330, Brier=0.327
G0: ECE=0.365, Brier=0.364
G1: ECE=0.387, Brier=0.376
G2: ECE=0.381, Brier=0.377
Observed correct fraction
Mean positive-class proxy score
0.0
0.2
0.4
0.6
0.8
1.0
Filled points: bin n>=10; small hollow points: n<10. All bins enter ECE. No fitting/recalibration. Bin counts in workbook.

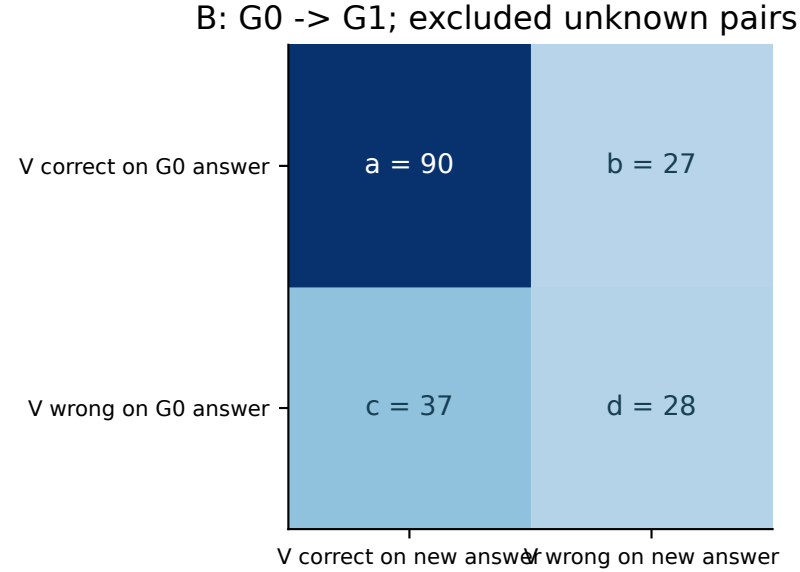

Coverage, generator correctness and verifier baselines
B: G correctness
Verifiable-subset accuracy
All-output identification bounds
B: coverage exclusions
Unknown / all
Truncated / all
B: V and baseline
V accuracy
V balanced accuracy
Always accept baseline
C: G correctness
Verifiable-subset accuracy
All-output identification bounds
C: coverage exclusions
Unknown / all
Truncated / all
C: V and baseline
V accuracy
V balanced accuracy
Always accept baseline
G0
G1
G2
Correctness shading: unknown-label identification bounds, NOT confidence intervals. Unknown labels remain unknown; truncation is a subset of unknown.

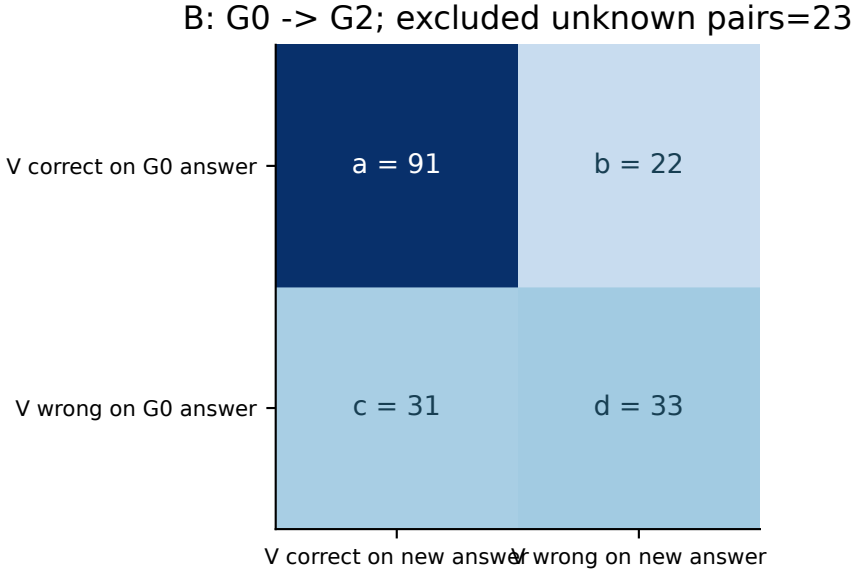

Exploratory length-stratified V0 errors
B: FNR by output length
B: FPR by output length
C: FNR by output length
C: FPR by output length
G0
G1
G2
<=512
513-1024
1025-2048
>2048
Output tokens
Small numbers above points = relevant true-class support. Missing denominator = NA. Strata can be sparse; length differences are not causal mechanisms.

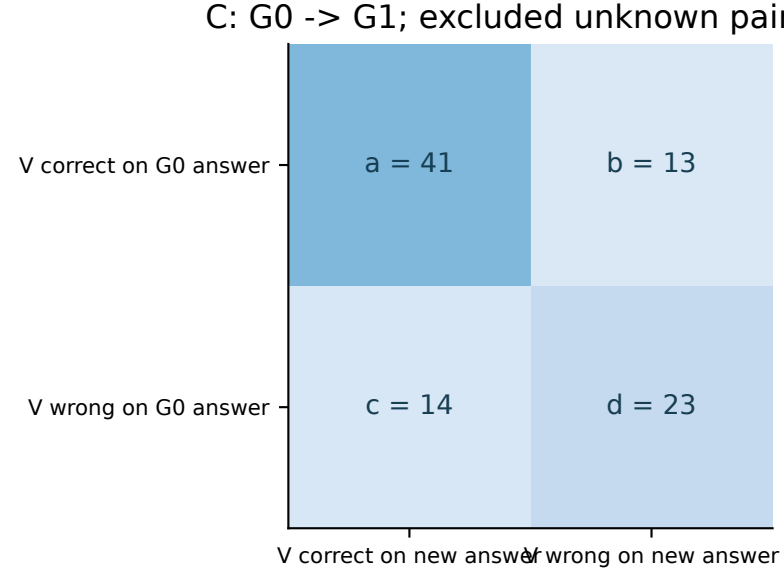


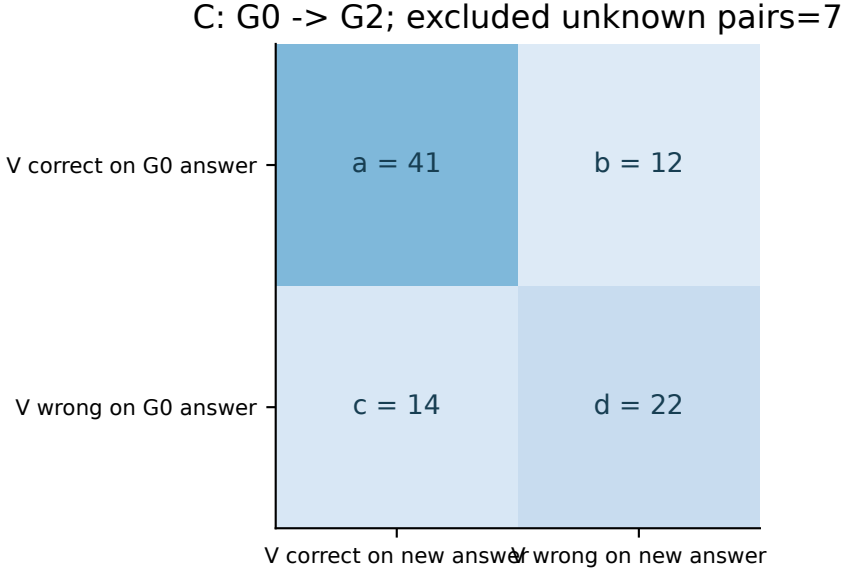
Shared 0-1 color scale. Grey = no relevant true-class examples. At most 4 candidates/question: local proportions are noisy. IDs in per_question_metrics.csv.

Descriptive paired-question judgement transitions

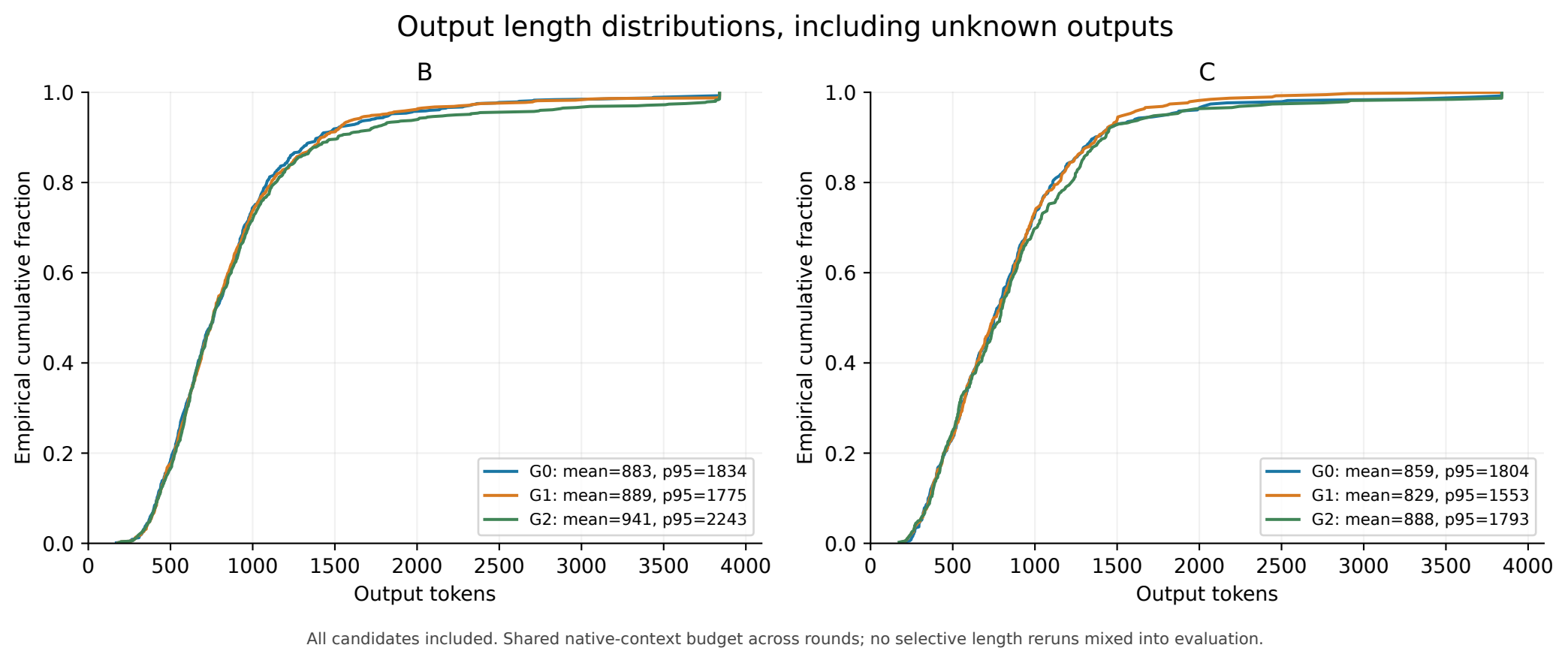


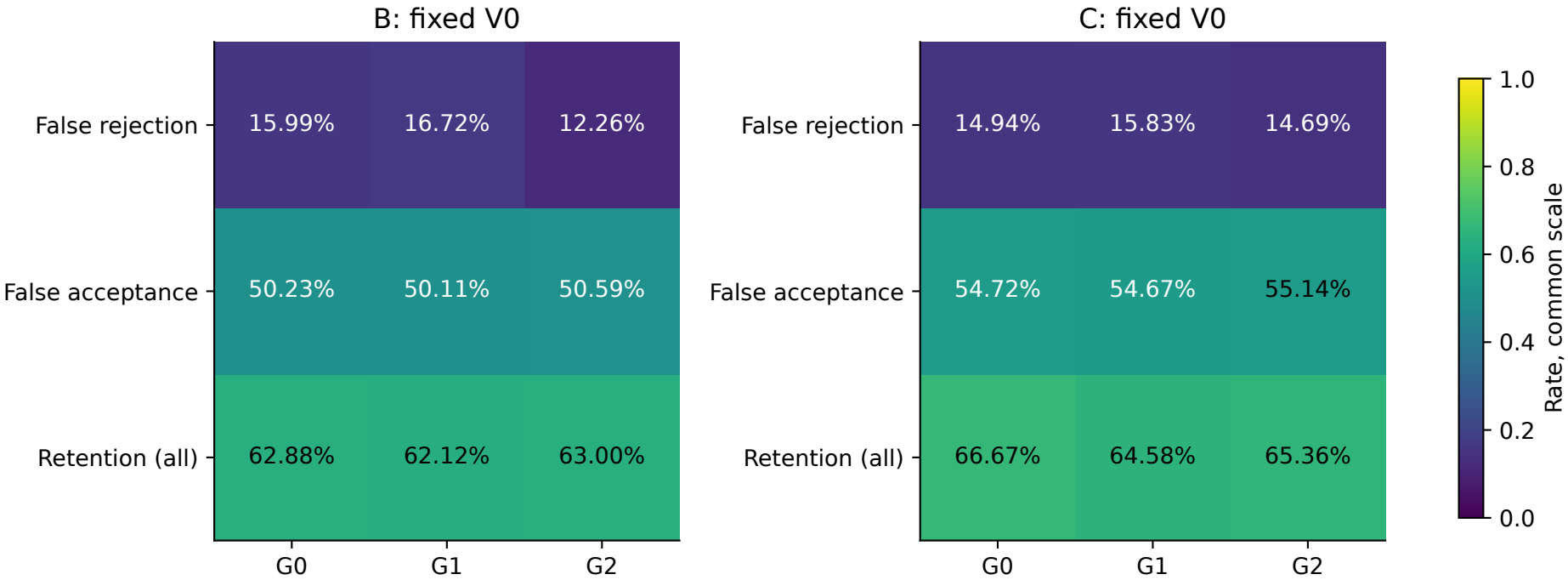






Candidate index 0 only. Different G answers and truth can occur. These are NOT fixed-answer verifier-forgetting matrices.


Output length distributions, including unknown outputs
B
C
Empirical cumulative fraction
Output tokens
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
2500
3000
3500
4000
G0: mean=883, p95=1834
G1: mean=889, p95=1775
G2: mean=941, p95=2243
G0: mean=859, p95=1804
G1: mean=829, p95=1553
G2: mean=888, p95=1793
All candidates included. Shared native-context budget across rounds; no selective length reruns mixed into evaluation.


Fixed V0 x G rounds: error rates and retention
B: fixed V0
C: fixed V0
False rejection
False acceptance
Retention (all)
15.99%
16.72%
12.26%
50.23%
50.11%
50.59%
62.88%
62.12%
63.00%
14.94%
15.83%
14.69%
54.72%
54.67%
55.14%
66.67%
64.58%
65.36%
G0
G1
G2
Rate, common scale
1.0
0.8
0.6
0.4
0.2
0.0
No Vrand or Vtarget checkpoints evaluated. All panels share the same 0-1 color scale; unknown excluded only from FNR/FPR.